

Apparent Probability Weighting in Mirrors: Choice Set Symmetry and Training*

Mark Dean and Hanyao Zhang

May 13, 2026

Abstract

Oprea (2024) demonstrated that deterministic ‘mirrors’ can also lead to the classical ‘four fold pattern of risk’ in experimental subjects, suggesting that such behavior may in part be driven by the complexity of stimuli. Recent critiques by Banki et al. (2025) and Wu (2025) have suggested that Oprea’s results may be driven by subject confusion, both about the nature of the task and whether mirrors are in fact lotteries. We present results from a new set of experiments that suggest that this is not the case. Building off the experiments of Wu, which were designed to eliminate these sources of confusion, we find robust evidence for the four fold pattern for mirrors. The difference in results is instead be driven by two other sources. First, the choices in Wu are designed to be symmetric around the expected value of the lottery, meaning that ‘constant’ errors do not contribute to the four fold pattern. It turns out this channel is important. Second, the training regime of Wu appears to teach subjects how to calculate expected values, rather than simply removing confusion.

1 Introduction

A vast experimental literature has demonstrated systematic violations of expected utility theory in lottery choice. One classic example is the so-called ‘four fold’ pattern of risk

*Department of Economics, Columbia University. Dean: mark.dean@columbia.edu. Zhang: hanyao.zhang@columbia.edu

(henceforth FFP), which can be interpreted as a lack of responsiveness in valuations to changes in probability parameters. Recent work by Oprea (2024, henceforth Oprea) found a similar pattern of behavior in valuations of deterministic ‘mirrors’ – choice alternatives that have a structure similar to lotteries, but which pay out deterministically and so have an objectively correct value. This has led people to consider the possibility that at least some of the FFP in lotteries is driven by a response to their complexity, rather than preferences for risk.

In response to Oprea, work by Banki et al. (2025, henceforth BSWW) and Wu (2025, henceforth Wu) has argued that the similarity between lottery and mirror valuations has been driven by two types of subject confusion – specifically (i) that Oprea’s instructions leave subjects confused in both mirror and lottery tasks, meaning they respond *mindlessly* and (ii) that subjects may incorrectly believe that deterministic mirrors are in fact lotteries. BSWW rely on a reinterpretation of the data from the original Oprea paper. Their main result is to show differences in behavior between subjects that accurately answered comprehension questions first time and those that did not, interpreting the former as the group that do not suffer from confusion. Wu describes a new experiment, which makes several changes from the design of Oprea aimed at reducing confusion, which they report as showing FFP for lotteries but not mirrors.

In this paper we argue that the main result of Oprea – FFP appearing in mirrors with strength close to lotteries – is not due to unusually high levels of subject confusion. We conduct new experiments, the design of which adheres closely to that of Wu, in which we find robust FFP in both lotteries and mirrors. We identify two non-confusion factors that lead to the discrepancy between Oprea and Wu. First, we show that the FFP remains in mirrors if the range of values that a subject can express is fixed across lotteries, rather than being symmetrically centered on the expected value of the lottery as it is in Wu. This is due to the presence of ‘constant’ (or trembling hand) errors — choice mistakes that are independent of the utility difference between alternatives — which are common to lotteries and mirrors and can cause FFP in a ‘fixed value’, but not a ‘centered value’ design. This observation helps to rationalize the finding of Harbaugh, Krause and Vesterlund (2010), that FFP is stronger in valuation tasks than binary choice tasks. Second, we provide evidence that Wu’s original

training works in part by teaching subjects how to calculate expected values, rather than merely reducing confusion about what the experiment is asking. In sum, we argue that Wu’s result differ from Oprea’s not because Wu’s design eliminates confusion, but because of other changes Wu makes to Oprea’s experimental design and statistical procedure.

Oprea’s original design involved lotteries and mirrors that varied in the probability p of winning a fixed prize. For lotteries, the subject would receive the fixed prize x with probability p , while for mirrors the subject would receive the expected value px for sure. Wu makes several changes to address potential sources of confusion in Oprea’s design, specifically introducing (a) a between subject design (b) a training stage in which simulated results of lotteries and mirrors are shown (c) comprehension questions that route subjects away from the experiment if too many fail (d) simplified lotteries (\$100 prize in Wu vs \$25 in Oprea) (e) binary choices rather than MPL valuation tasks and (f) the certain payments that make up the lottery choice are symmetric around the expected value of each lottery.

Given that all these changes are made at once it is hard to know which is responsible for the differential results between the two papers. Our hypothesis is twofold. First, that the change from an MPL fixed across lotteries to certain prospects which are centered on the expected value of each lottery effects the range of behaviors that can be expressed. This eliminates a particular channel through which FFP may operate – constant errors. As we show in section 2.1, if subjects are faced with an MPL or set of binary choices with certain payments which are invariant across lotteries, constant errors can lead to big deviations from expected value maximization when probabilities are high or low, leading to the FFP.¹ Oprea uses fixed MPLs, as does much of the experimental literature studying FFP. In contrast, if subjects are asked to make a set of choices which are centered on the EV of each lottery, as in Wu, then constant errors will average out to zero.

Our second hypothesis is that the training regime of Wu might be implicitly teaching subjects how to calculate the expected value of a mirror, rather than simply reducing confusion. Both training and choices involved lotteries with a prize of \$100. If you are told that the amount you will get paid from an 80% chance of \$100 is \$80, \$80, \$80, \$80..., and a 60%

¹A related explanation for FFP can be found in Blavatsky (2007), which proposes that errors cannot lead to a lottery being valued above its highest or below its lowest prize.

chance of \$100 is \$60, \$60, \$60, \$60... it may become obvious how to translate p into values in this particular problem, even if you do not understand how to calculate expected values more generally.² Arguably, this goes beyond removing confusion about the payoff rule, and helps to inform people how to answer the question. Moreover, one might expect to be stronger for mirrors than lotteries, where the training feedback is stochastic.

In order to test our two hypotheses we run a set of new experiments using the online platform Prolific. Our baseline set up mimics that of Wu by making use of his training stage and comprehension questions (points (b) and (c) above). For efficiency reasons we run a within subjects design (point (a)), but we show our results are robust to analyzing only the first round for each subject, mimicking a between subject design. We use a set of choices that have common components across lotteries (point (f)).

Within this set up we run three experiments. The first (Experiment 1) asks subjects to value lotteries using an MPL. Subjects take part in a training regime which uses \$100 prizes, and are randomized into one of two treatments. The \$100 treatment also uses a \$100 prize in the experimental decisions, while the \$25 treatment uses a \$25 prize. The second experiment (the Binary Choice experiment) replicates the design of Experiment 1 but subjects face binary choices rather than MPLs (addressing point (e)). The third (the Alternative Training experiment) replicates the design of Experiment 1 but replaces the \$100 prize in the training regime with a \$50 prize.

This design allow us to test both our hypotheses. To test our first — that constant errors can contribute to the FFP when choice options are asymmetric — we can estimate FFP on our full data, and on a restricted data set which only uses only choices with symmetric certain values. To test our second — that Wu’s training can help subjects locally learn the expected value of mirrors — we can compare FFP across training regimes and prize levels in Experiments 1 and the Alternative Training experiment. if Wu’s design is simply clearing up confusion about the payoff rule, then it should work to reduce the four fold pattern in choice problems which have a different x between training and actual tasks. In contrast, if subjects

²Indeed this issue is anticipated in Oprea (2024), which states “To give an extreme example, if we were to directly tell subjects in our experiment the expected value, we would expect the classical pattern to shrink much more in mirrors than in lotteries. This is because knowing the expected value directly removes the difficulties of valuation in mirrors (since the value of mirrors just is the expected value), but it doesn’t in any clear way remove those same difficulties in lotteries.” (p. 3805 footnote 8)

are mechanically learning about the mapping from p to value in one environment then they may still exhibit the four fold pattern in a different environment.

We organize our results as follows. In section 4 we show that the training regime and comprehension questions of Wu do not, on their own, reverse the result of Oprea. When we pair these design features with the \$25 prizes and MPL design from Oprea’s original paper (the \$25 treatment in Experiment 1), we find strong and consistent FFP behavior in both mirrors and lotteries. The degree of FFP is correlated across the two stimuli. Furthermore, we show that neither a between subject design or a switch to binary choices reverses Oprea’s findings.

In section 5 we argue that the difference between the two results is driven at least in part by the symmetry or otherwise of questions around the expected value of the lottery. Again using Experiment 1, \$25 treatment, we replicate the analysis of Wu by treating each line in the MPL as a binary choice question, then fitting a logistic regression. We show that, using all the questions from the MPL, we continue to find FFP behavior for both lotteries and mirrors. However, if we restrict our analysis to questions which are selected such that the certain options are symmetrically distributed around the expected value of the lottery, then the FFP weakens significantly, and more so for mirrors than lotteries. A similar pattern obtains – although more noisily – if we replace data from our MPL experiment with that from our binary choice experiment.

In clarifying the source of difference between Oprea and Wu’s results, we identify two distinct causes of FFP-type behavior. The first are ‘constant’ (or trembling hand) errors: choice mistakes that are independent of the utility difference between alternatives. Such errors arise in both lotteries and mirrors, and have been heavily studied in the lottery choice literature (see for example Harless and Camerer (1994), Wakker, Erev and Weber (1994), Loomes, Moffatt and Sugden (2002)). The second are smaller errors centered around the expected value, similar to Fechner errors. These arise mostly in lotteries, not in mirrors.

These two different types of errors explain the different results between the full (asymmetric) and restricted (symmetric) data sets. Because both lotteries and mirrors exhibit constant errors, choices are very similar between the two when the certain value is a long way from the expected value of the prospect. In both cases a stable fraction of subjects

make the ‘wrong’ choice regardless of how far the certain option is from the expected value. When the logistic regression is fit over the entire sample, these observations carry a lot of weight, meaning that estimates are similar between lotteries and mirrors. In contrast, when close to the expected value, there are significant differences between lotteries and mirrors. In particular significantly more people switch exactly at the expected value for mirrors than they do for lotteries. When fit only to the symmetric set of questions with options close to the expected value, it is this pattern that drives the results, leading to different estimates between lotteries and mirrors.

There are of course different interpretations of the constant errors: they may represent a subject’s failure to be able to calculate what they want to calculate, or a failure to understand how to best respond to the incentive structure of the experiment. Whether we refer to the latter as ‘confusion’ may be a matter of taste. However, we hope that by adopting the the training and comprehension check regime of Wu we have largely removed any confusion caused by ambiguity or low quality training.

Further, we argue that constant errors are common to many lottery valuation experiments, and therefore (i) not an artifact of our (or Oprea’s) instructions and (ii) an important part of understanding FFP *in lotteries*. For example, for lotteries with a 10% probability of winning a positive prize (and zero otherwise), in our data subjects report a certain equivalence above 50% of the prize value 19% of the time and above 80% of the prize 9% of the time. The equivalent figures are 13% (8%) for Gonzalez and Wu (1999), 13% (6%) for Choi et al. (2022)³ and 19% (7%) in Enke and Graeber (2023).⁴ Indeed there exists a long literature showing how valuations for lotteries can be biased towards the center of the an MPL on average⁵ for which constant errors are one possible cause.

Our results so far have established that the difference between Wu and Oprea’s result are not driven by a reduction in confusion, but largely differences in the way the choice sets are constructed in the two experiments. Finally, in section 6, we show that, as per our second hypothesis, the training regime matters.

³This paper does not use a 10% lottery so instead we report results for the 5% lottery

⁴This refers to the MPL version of their experiment without added complexity or cognitive load. For the BDM version the numbers are 10% and 4%.

⁵See of example Birnbaum (1992), Harrison et al. (2005), Beauchamp et al. (2020).

In Experiment 1 we find that subjects on average exhibit more FFP behavior for mirrors in the \$25 prize treatment than they do in the \$100 prize treatment. This could be driven by the exposure to a similar prize in training and experimental questions, or the fact that subjects simply find it harder to calculate expected values for prizes other than \$100.⁶ To differentiate between the two possibilities we use the \$50 the training regime of the Alternative Training experiment. This causes the difference in responsiveness to mirrors in the \$100 and \$25 prize regimes to disappear, indicating the importance of training. While our design does not allow for a clean comparison between the two training regimes, if anything it appears that the \$50 training leads to more responsiveness in both lotteries and mirrors.

2 Conceptual Framework

Let $m_{p,x}$ and $l_{p,x}$ be, respectively, binary mirrors and lotteries with probability p of winning a fixed prize x (and 0 otherwise). Let $v^i(m_{p,x})$ be the stated value of a mirror for subject i , assuming that the DM recognizes it as a mirror. We assume that

$$v(m_{p,x}) = px + \varepsilon_{calc}^i + \varepsilon_{conf}^i$$

where ε_{calc}^i is an error caused by a failure to perform the relevant calculation px , or a failure to respond correctly to incentives that they otherwise understood. ε_{conf}^i is an error term caused by confusion in the instructions about what it is the subject is being asked to do - i.e. if the subject fully understood the incentive structure in the task then ε_{conf}^i would be zero. We assume that ε_{calc}^i and ε_{conf}^i are stochastic, with mean zero and variances $\sigma_{\varepsilon,calc}^2$ and $\sigma_{\varepsilon,conf}^2$ respectively. Note that generically, for a Bayesian agent, increases in the variance of ε_{calc}^i and ε_{conf}^i will reduce the responsiveness of $v(m_p)$ to p and so increase the four fold pattern of risk.⁷

⁶Arguably, this would be in line with Oprea's original explanation.

⁷For example, if we assume that the error terms are normal and that the DM has a normal prior for the value of a mirror $\bar{v}$ with variance σ_v^2 then mean valuation for $m_{p,x}$ will be

$$\lambda px + (1 - \lambda)\bar{v}$$

Let $v^i(l_{p,x})$ be the stated value of a lottery $l_{p,x}$. We assume that

$$v^i(l_{p,x}) = u^i(l_{p,x}) + \mu_u^i + \frac{\mu_{calc}^i}{1 + \mu_u^{i2}} + \mu_{conf}^i$$

where $u^i(l_{p,x})$ is the true utility of $l_{p,x}$ to the subject. μ_u^i is the uncertainty about that utility - i.e. uncertainty about how to value a lottery which is unrelated to any ability to carry out calculations. Implicitly we can think of mirror valuations as having an equivalent noise term which is set equal to zero. The term $\frac{\mu_{calc}^i}{1 + \mu_u^{i2}}$ is designed to capture that fact that, if the subject is very uncertain about their true utility, calculation errors are relatively unimportant and reducing them will do little to reduce decision noise.

We also allow for the possibility that a subject is confused between lotteries and mirrors. Let $v^i(m_{p,x})$ be the true stated valuation for the mirror p then

$$v^i(m_{p,x}) = I^i v^i(m_{p,x}) + (1 - I^i) v^i(l_{p,x})$$

Where I is an indicator function which takes the value 1 if the subject correctly understands a mirror as a mirror and 0 otherwise.

Within this framework, our interpretation of Oprea's original argument is that the variance of ε_{calc}^i is highly correlated with the variance in μ_{calc}^i in the population, and this is the primary driver of the four fold pattern of risk for both lotteries and mirrors. This explains both the existence of the four fold pattern in the mirrors and the between-subject correlation between the four fold pattern in mirrors and lotteries. A more extreme version of this argument would set $u^i(l_{p,x}) = px$ and $\mu_u^i = 0$, essentially saying that calculation errors are identical between lotteries and mirrors, and are the only source of noise in lottery choice

In contrast, we interpret BSWW as claiming that $I^i = 0$ for a significant fraction of the population, and/or that ε_{conf}^i is high, and highly correlated with μ_{conf}^i . Both of these claims would indeed explain the findings in the original Oprea experiment. Moreover, they interpret

where

$$\lambda = \frac{\sigma_v^2}{\sigma_v^2 + \sigma_{\varepsilon, calc}^2 + \sigma_{\varepsilon, conf}^2}$$

And so the DM will be risk averse for mirrors with values above $\bar{v}$ and risk loving for mirrors with values below $\bar{v}$.

the experiment in Wu as correcting both these features (i.e. setting $I^i = 0$ for most subjects and reducing the variance of ε_{conf}^i and μ_{conf}^i). The finding that $v(m_{p,x}) = px$ for many subjects is then taken as proof that ε_{calc}^i is essentially unimportant, both in their experiment and Oprea's original study.

2.1 Constant Errors and Asymmetric Valuations

We next show how constant errors can lead to FFP in a set up where the same set of possible valuations can be expressed for each mirror, but not if the set of possible valuations is centered on the EV of each lottery.⁸ Consider a simplified version of the set up above in which

$$v(m_{p,x}) = px + \varepsilon$$

Assume that the DM can pick from a continuum of possible values between $v_*(xp)$ and $v^*(xp)$ which may depend on the underlying problem x, p . Assume that ε has a 'constant' structure, such that the DM makes no errors with probability q but with probability $(1 - q)$ they pick randomly from $[v_*(xp), v^*(xp)]$ (Loomes, Moffatt and Sugden, 2002). Formally,

$$\varepsilon = \begin{cases} 0 & \text{with probability } q \\ U[v_*(xp), v^*(xp)] - xp & \text{with probability } 1 - q \end{cases}$$

where U is the uniform distribution. Then, for the problem xp

$$E(v(m_{p,x})) = qpx + (1 - q) \frac{v^*(xp) + v_*(xp)}{2}$$

In the Oprea case,⁹ $v^*(xp)$ and $v_*(xp)$ are fixed such that $\frac{v^*(xp) + v_*(xp)}{2} = 12.5$. Thus, for $px < 12.5$, $E(v(m_{p,x})) > px$, while for $px > 12.5$, $E(v(m_{p,x})) < px$ - i.e. the FFP. In contrast, in Wu, effectively $v^*(xp)$ and $v_*(xp)$ are chosen such that $\frac{v^*(xp) - v_*(xp)}{2} = xp$, and so $E(v(m_{p,x})) = px$ for every px .

Note that constant errors are one possible cause of such an effect, but any force which

⁸For simplicity we model a valuation task. A similar argument applies to binary choice tasks.

⁹We only consider the case of gains, and the case of losses follows a similar logic.

pushes subjects towards the center of an MPL will also cause FFP in with asymmetric but not symmetric valuations. A significant literature has documented the importance of such compromise effects (Birnbaum, 1992, Harrison et al., 2005, Beauchamp et al., 2020).

3 Experimental Design

All our experiments are based on binary lotteries $(\$x, p; \$0)$ (with p chance winning $\$x$, and $\$0$ otherwise). Each subject faces five different p in the experiment: $p \in \{10, 25, 50, 75, 90\}$, and is randomized into one of the two prizes: $\$25$ and $\$100$. The prize stays the same for a fixed subject throughout the experiment. In the same experimental session, subjects are also asked to value the deterministic mirrors of each of the five lotteries.

In our experiments, a lottery $l_{p,x}$ and its deterministic mirror $m_{p,x}$ are both presented as 100 boxes, among which $100p$ contain $\$x$ and $100(1 - p)$ contains $\$0$. Although lottery and mirror share exactly the same visual presentation, they differ in how the payment is determined from the boxes. The realized payout of a lottery is determined by the amount in a randomly drawn box, while the mirror pays out the average amount of money contained in these 100 boxes with certainty.

All our experiments are split into two payment rule blocks – one containing all lottery tasks and the other containing all mirror tasks. The order of blocks is randomized at the subject level. Since we do not inform the subjects of the second payment rule when they are performing tasks with the first, we are always able to drop the second payment rule of each subject and revert to the between-subject payment rule treatment in Wu.¹⁰

All our experiments adopt Wu’s experimental instructions, subject training, and comprehension checks. However, one key point of difference between our paper and Wu is that the amount used in the training and comprehension check, $\$y$, may differ from the prize $\$x$. Before each payment rule block, our subjects go through three rounds of training. In each training round, the subjects are presented with a set of 100 boxes, and are shown 10 simulated outcomes according to the payment rule in this block. For example, if the block

¹⁰Contrary to BSWW, the results turn out to be similar regardless of which payment rule comes first. Thus, we focus on our full data set in this paper and relegate the between-subject payment rule results to the appendix.

involves lotteries, and $l_{p,y}$ is involved in the simulation, 10 independent realizations will be sampled using $l_{p,y}$. On the other hand, if the block involves mirrors, and $m_{p,y}$ is involved in the simulation, the certain payout $\$py$ will be showed 10 times. After the training, the subjects need to pass the comprehension check before moving to the actual tasks. There are five questions in the comprehension check, and the subject has to correctly answer four to pass. Subjects who fail more than 1 questions are screened out from the experiment, and are not included in our data. An example question is shown in Figure 1. Each question in the comprehension check gives the subjects 100 boxes, among which p contain $\$y$ and $100 - p$ contain $\$0$. The question asks the subject to select all possible payout from these 100 boxes, according to the current block’s payment rule. If the current payment rule is lottery, the subjects should select both $\$0$ and $\$y$, and if the current payment rule is mirror, the subjects should select $\$py$ only.

Imagine you have the following option. If we simulate this option just like what we did, what possible outcomes might you see? Choose all that apply.

[Review Instructions](#)

30 boxes	70 boxes
\$50	\$0

☐	☐	☐	☐	☐	☐	☐
\$0	\$5	\$15	\$25	\$35	\$45	\$50

Submit

Figure 1: An example comprehension check

We conduct three experiments in total. Our main experiment (Experiment 1) follows Oprea’s experimental design, where we elicit valuations of lotteries and mirrors using multiple price list (MPL), with enforced single switching. Since Wu’s experimental instructions do not involve MPL, we adopt the MPL instructions from McGranahan et al (2024). The amount used in training is \$100, the same as in Wu. The price list we use has 26 rows with equal increments – under prize \$25, subjects choose between a lottery/mirror and 26 certain payments ranging from \$0 to \$25, with increments of \$1; under prize \$100, subjects choose between a lottery/mirror and 26 certain payments ranging from \$0 to \$100, with increments

of \$4. We enforce a unique switching point, and define the valuation as the midpoint of the two certain payments at the switching rows.

We also conduct two additional experiments. The first is the Binary Choice experiment. Each subjects make 10 binary choices, randomly selected from 40 candidate binary choice tasks for their prize, as shown in Table 1. The amount used in training is \$100, the same as in Wu. The experiment is almost the same as Wu’s experiment except for two aspects. First and most crucially, while Wu only includes binary choices such that the certain payment is symmetric around the expected value of the lottery/mirror, and are not too far away from the expected value, we included binary choices that involve asymmetric certain payments. For example, for choices involving the lottery $l_{75,100}$ (expected value = \$75), the smallest certain payment that Wu asks subjects to choose against the lottery is \$60, and the largest is \$90. In contrast, we ask subjects to choose between $l_{75,100}$ and certain payment as small as \$4, or as large as \$96. This design choice is to test our hypothesized effects of choice set symmetry. Second, as in all of our experiments, we let the same set of subjects value both lotteries and mirrors in two separate blocks.

p	certain payments (\$) (Prize \$25)	certain payments (\$) (Prize \$100)
10	1, 2, 3, 4, 10, 15, 21, 24	4, 8, 12, 16, 40, 60, 84, 96
25	1, 4, 6, 7, 10, 15, 21, 24	4, 16, 24, 28, 40, 60, 84, 96
50	1, 4, 10, 12, 13, 15, 21, 24	4, 16, 40, 48, 52, 60, 84, 96
75	1, 4, 10, 15, 18, 19, 21, 24	4, 16, 40, 60, 72, 76, 84, 96
90	1, 4, 10, 15, 21, 22, 23, 24	4, 16, 40, 60, 84, 88, 92, 96

Table 1: All choices tasks in the experiment

In the second additional experiment (Alternative Training experiment), we keep all the design choices in the main experiment with only one exception: We use \$50 as the amount in the training and comprehension check. This experiment is to test our theory about Wu’s training teaching more than just the distinctions between lottery and mirror, but also involving mechanically memorizing patterns.

The Experiment 1 was conducted on Prolific in May 2025. The Binary Choice and the

Alternative Training experiment were conducted on Prolific in June and September 2025, respectively. Each subject received \$3 payment upon completion. If a subject is screened out in their second payment rule block, they are paid \$1.5 instead. Table 2 shows the number of participants who started each experiment (provided their informed consent) and who finished each experiment. It also shows the fraction of started subjects who finished the experiment, who failed the comprehension check in the first payment rule, and who failed the comprehension check in the second payment rule. The three fractions do not add to 100% since some subjects withdrew in other parts of the experiment. The fractions of people who failed the first comprehension check is comparable to the number in Wu (2025) (13.6%), though we end up excluding more subjects because we have two separate comprehension checks for the two payment rules.

Experiment	Started	Finished	Finished (%)	Failed 1st (%)	Failed 2nd (%)
Main	571	292	51.1%	15.4%	22.1%
Alternative training	545	273	50.1%	16.7%	24.4%
Binary choice	548	295	53.8%	18.1%	13.1%

Table 2: Participant completion and comprehension check failure in the experiments

We incentivize the responses by the following mechanism: For a subject who completes both blocks, at the end of the experiment, with a probability $1/X$, where X is the subject-specific prize, one randomly chosen task determines their bonus payment. Complete instructions and experimental screenshots are included in the appendix.

4 Oprea’s Conclusions Survive in ‘Well-Trained’ Subjects

In this section, we replicate Oprea’s original analysis using the data set from the main experiment. To maximize comparability with Oprea, we focus on the \$25 treatment from Experiment 1.

Our Figure 2 re-establishes Oprea’s main result. As in Oprea’s Figure 1 (pg. 3796), on the x -axis, we plot the probability of winning the prize, and on the y -axis, we show the mean deviation of the valuations from expected value. In both Lottery and Mirror treatments, the valuations exhibit the classic probability weighting pattern – overvaluing small probability gains and undervaluing high probability gains. This is true even when mirrors entail no risk. Moreover, in the Mirror treatment, the probability weighting is almost as strong as in the Lottery treatment.

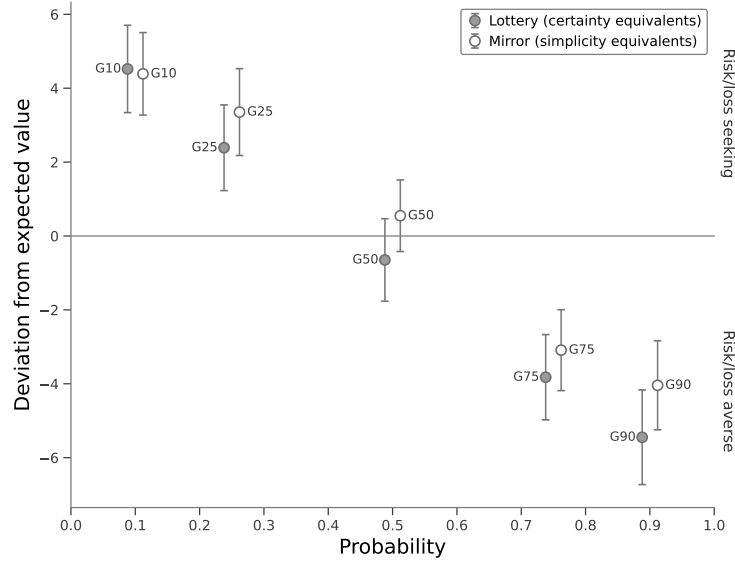


Figure 2: Mean Deviations from Expected Value in Lotteries (Solid Gray Dots) and Mirrors (Hollow Dots)

Figure 3 replicates Oprea’s Figure 4 (pg. 3801), which examines the correlation between a subject’s deviation from expected value in the Lottery and Mirror treatments. The left panel shows this correlation for the absolute deviations. Each scatter is an individual subject, and the x -axis is that subject’s mean absolute deviation from the expected value-maximizing choice in mirrors and the y -axis plotting the same deviation in lotteries.

The right panel differs from the left by plotting the normalized deviations, defined as

$$\text{normalized deviation} = \text{sign}(\text{fourfold pattern}) \times (\text{valuation} - \text{EV}), \quad (1)$$

where $\text{sign}(\text{fourfold pattern}) = -1$ if $p \geq 0.5$, and $\text{sign}(\text{fourfold pattern}) = 1$ otherwise. This

sign is the direction of deviation predicted by fourfold pattern. Therefore, the normalized deviation has the interpretation as the strength of the fourfold pattern.

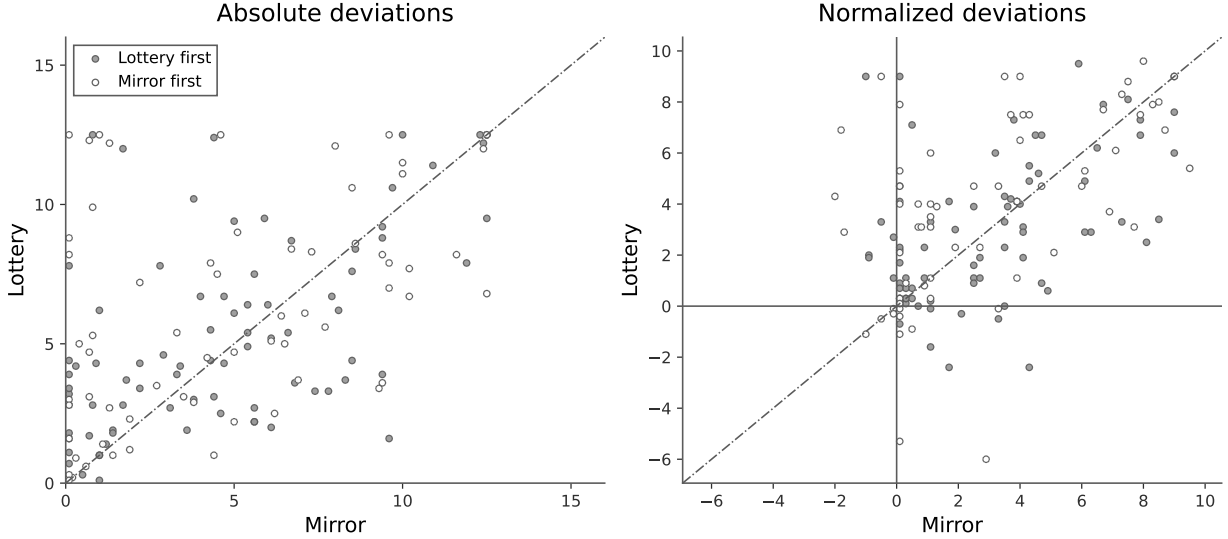


Figure 3: Deviations from Expected Value-Maximizing Choices in Mirrors (x-axis) versus Lotteries (y-axis), by Subject

Figure 3 re-establishes Oprea’s findings on this correlation. First, the Pearson coefficient of correlation is 0.50 for the absolute deviations, and 0.51 for the normalized deviations, showing a strong correlation between lottery and mirror deviations.¹¹ Second, in the right panel, most of the scatters are in the first quadrant, showing deviations consistent with probability weighting in both Lottery and Mirror treatments.

Since one main point of difference between Oprea and BSWW is the choice of summary statistic, it is also worth it to go beyond the mean and closely examine the complete distribution of responses. Viewing the MPL as a table of binary choices, Figure 4 shows the fraction of subjects that prefer the certain payment, for each binary choice between a lottery/mirror and a certain payment. Each panel shows for a fixed lottery/mirror, the share of subjects who prefer the certain payment in the y -axis as a function of the certain payment in the x -axis. The expected value is marked with the red vertical line. As documented in Oprea’s reply to BSWW (Figure 2 in Oprea, 2025), we find that for both lottery and mirror, the probability mass that is consistent with the FFP is much larger than that is

¹¹For comparison, in Oprea’s original data, this correlation is 0.68 for the absolute deviations and 0.62 for the normalized deviations.

inconsistent. For example, facing $l_{0.1,x}$ ($m_{0.1,x}$) with EV \$2.5, 59.0% (50.0%) of all subjects choose the lottery (mirror) over a certain payment of \$3, consistent with FFP. However, only 29.9% (15.2%) of all subjects choose a certain payment of \$2 over the lottery (mirror), to the opposite of FFP. Moreover, the primary difference between lottery and mirror CDFs are around the expected values, but the CDFs tend to be similar away from the expected values. We will return to this observation in Section 5.

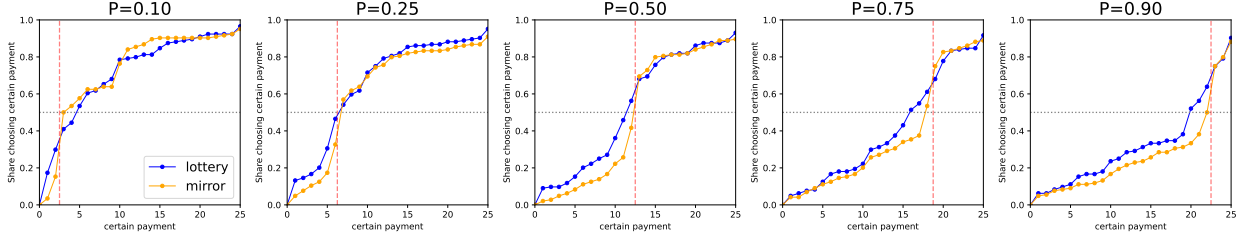


Figure 4: CDF of Valuations

We also directly examine one of BSWW’s key claims that Oprea’s results are driven by an unusually high fraction of subjects who respond mindlessly. One particularly strong indication of mindless responding is the violation of first order stochastic dominance (FOSD): In other words, some subjects do not give higher valuations to lotteries with higher probability to win. Table 3 compares the rate of FOSD violation in our main experiment with Oprea. Following BSWW, we detect FOSD violation by running the regression

$$\text{valuation} = \beta_0 + \beta_1 \times p,$$

for each subject, separately for lottery and mirror. If the *responsiveness* β_1 is strictly smaller than zero, we say this subject violates FOSD.

In the main experiment under prize \$100, the average responsiveness is 0.502 for lottery and 0.552 for mirror. Table 3 shows the fraction of subjects who violates FOSD in the main experiment and in Oprea. The main experiment does achieve much lower rate of FOSD violations due to extensive training and subject screening. However, as we have established previously in this Section, this decrease of FOSD violation does not translate into the invalidation of Oprea’s core claim.

	Oprea (2024)’s main experiment	Main experiment, Prize \$25
Lottery	19.57%	8.24%
Mirror	22.28%	7.16%

Table 3: Fraction of subjects who have responsiveness **strictly** smaller than 0

Notes: To better compare with this paper, only the lotteries with a positive prize (the “G” lotteries) are included in Oprea (2024).

At this stage we have ruled out (b) the existence of a training stage and (c) comprehension checks that route subjects away from the experiment as the driver of the difference between Oprea and Wu’s results. In Appendix A we show that neither (e) binary choices nor (a) a between-subject design are responsible. We show the former by estimating mean valuations from our Binary Choice experiment, and the latter by replicating all our results while restricting the analysis to the first payment rule that each subject faced, which is insulated from any possible interference from the second payment rule. In Section 5 we show that (f) —symmetry of certain payments — plays a big role in the difference in results, while in section 6 we further examine the role of (b) the training regime and (d) simplified lotteries.

5 Choice Set Symmetry Reconciles Oprea and Wu

The key point of controversy between Oprea and Wu lies in whether mirrors exhibit the probability weighting patterns with strength broadly comparable to that in lotteries. As we hypothesize in Section 2.1, constant errors can lead to patterns akin to probability weighting when the range of expressible choices is asymmetric around the expected value of the lottery. Since this asymmetry is built into the MPL employed in Oprea (and most other studies of the FFP that use either MPLs or BDM valuations), but is absent in Wu’s list of binary choices, it may reconcile the seemingly contradictory findings in Oprea and Wu.

To test this hypothesis in our main treatment, we compare the valuations estimated from the standard MPL and the *symmetrized* MPL. We view the standard MPL as a list of binary choices in this data analysis. For example, the MPL involving the lottery $l_{p,25}$ can be decomposed into the 26 binary choices between $l_{p,25}$ and \$0, \$1, $\dots$, \$25. To symmetrize an

MPL, we only consider the four binary choices that are closest to the expected value. For example, for the lottery $l_{90,25}$ (EV=\$22.5), we consider the four binary choices between the lottery and \$21, \$22, \$23, \$24, and discard the binary choices between it and, for example, \$15, to achieve symmetry. In other words, we test our hypothesis using two nested data sets collected in a single experimental interface, where the symmetrized MPL is simply a subset of the binary choices contained in the standard MPL.

We estimate the valuations by fitting logistic regressions on the binary choices in the standard MPL or the symmetrized MPL. This estimation procedure is exactly the same as that used in Wu. We first break down each complete or symmetrized MPL to a list of binary choices. Then, for each lottery and each mirror, we estimate the following model using the binary choices

$$P(\text{the lottery/mirror is chosen over } \$x) = f(\alpha + \beta x),$$

where $f(\cdot)$ is the standard logistic (sigmoid) function. The valuation x^* of this lottery or mirror is then defined implicitly by $f(\hat{\alpha} + \hat{\beta}x^*) = 0.5$. Note that although the standard MPL allows for a tight interval identification of the valuation, we deviate from this standard approach, and use the logistic regression approach to facilitate comparisons with the symmetrized MPL, where the standard approach is no longer applicable.

For brevity, we only illustrate the results for prize \$25 in this section. Figure 5 shows the recovered valuation of each lottery and mirror, along with the bootstrap 95% confidence interval. Under standard MPL, consistent with Figure 2, we replicate the classic probability weighting patterns in both lottery and mirror using this logistic regression method. Moreover, the strength of probability weighting is broadly comparable between lottery and mirror. The average normalized deviation (defined in Equation (1)) across the five lotteries is \$2.64, and across the five mirrors is \$2.05. In contrast, under symmetrized MPL, the probability weighting patterns become significantly weaker than under standard MPL, with average normalized deviation \$1.55 and \$0.75, for lotteries and mirrors respectively. This pattern validates our hypothesis over the relationship between symmetry and probability weighting. In addition, in the symmetrized MPL, probability weighting becomes visibly weaker in mirrors than in lotteries.

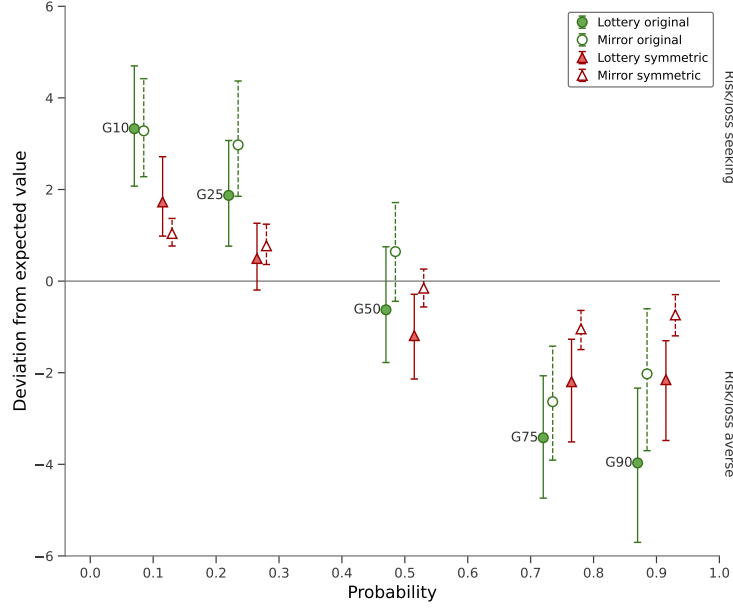


Figure 5: Logistic regression-estimated valuations, standard MPL versus symmetrized MPL
Note: The confidence intervals are constructed by subject-level bootstrap with 1000 replications.

To explain how choice set symmetry affects the valuation estimates and the lottery-mirror discrepancy, we look more closely at the distributions of choices subjects make. From Figure 4, two key observations arise. First, as pointed out by BSWW and Wu, the fraction of subjects preferring the certain amount rises sharply in mirrors around their certain values, but more gradually in lotteries around their expected values. Second, however, the lottery and mirror choice shares tend to converge when the certain amount is far away from the expected values. For example, when choosing between lottery $l_{10,25}$ and a certain payment of \$20, 9.0% of the 144 subjects choose the lottery over \$20, while in the choice between mirror $m_{10,25}$ and \$20, a similar fraction of 9.7% choose the mirror. In other words, a 9-10% of subjects state a preference for the lottery/mirror, even when the expected value of the lottery or the certain value of the mirror is much smaller than the certain payment. We refer to these choices as the constant errors.

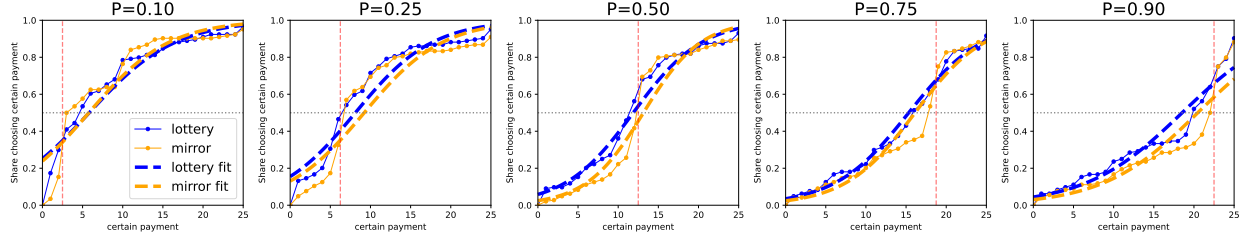


Figure 6: Choice shares in lottery and mirror in the MPL treatment, along with the logistic regression fit using the standard MPL data

Figure 6 overlays the logistic regression fits using the standard MPL data in dashed lines to Figure 4. These fits generally track the choice shares accurately in the regions far away from the expected value, but fail to accurately characterize the sharp increase of choice shares around the expected value. Since lottery and mirror choice shares converge in the regions away from the expected value, this explains why the valuation estimates are similar across lottery and mirror when using standard MPL data.

In contrast, Figure 7 shows the same series of choice shares and the logistic regression fit using the symmetrized MPL data (the shadowed region). The results are the exact opposite of Figure 6: the fits fail to accurately characterize the regions far away from the expected value by predicting more extreme probabilities than the actual data. On the flip side, the fits mechanically captures the sharp increase around the expected value. Since the region around the expected value are where mirrors differ from lotteries the most, this explains why the valuation estimates become quite different across lottery and mirror when using symmetrized MPL data.

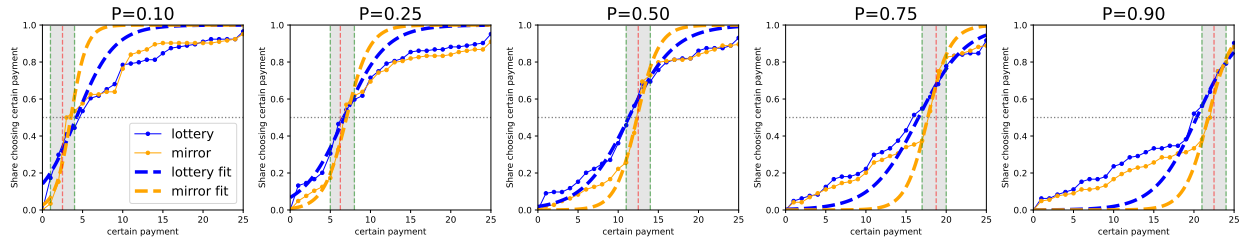


Figure 7: Choice shares in lottery and mirror in the MPL treatment, along with the logistic regression fit using only the symmetrized MPL data

We also briefly discuss the data from the Binary Choice experiment, where asymmetric choice sets again lead to probability weighting patterns. We symmetrize the binary choice

data set by only keeping the four binary choices closest to the expected value. We focus on mirrors here since lottery valuations cannot be precisely estimated with our sample size using logistic regression, especially with the symmetrized data set.¹² As can be seen in Figure 8, mirrors exhibit substantial probability weighting patterns in the complete data set, with average normalized deviation \$1.74. However, the valuations become close to their actual values in the symmetric data set, with average normalized deviation only \$0.39.

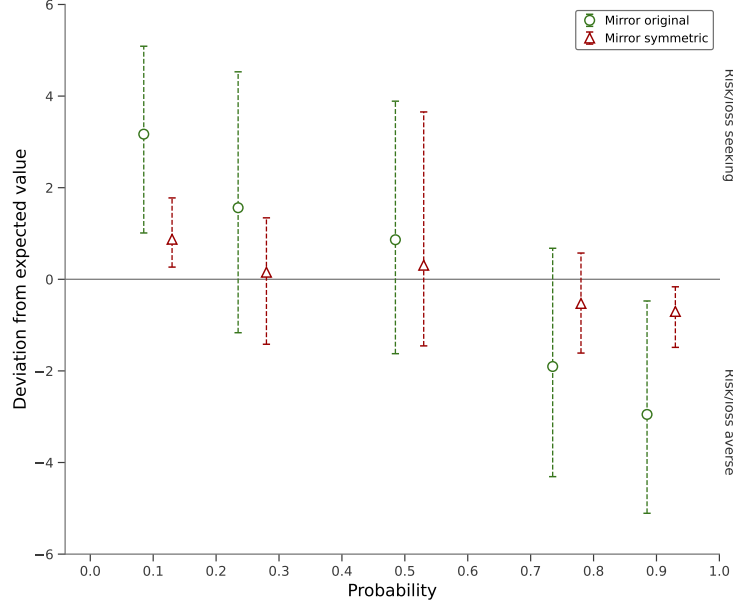


Figure 8: Logistic regression-estimated valuations, complete versus symmetrized choice set
Note: The confidence intervals are constructed by subject-level bootstrap with 1000 replications.

Choice set symmetry has strong impact on the valuation estimates due to the fact that the constant errors are not well accounted for in the econometric techniques that estimate valuations from observed choices. Although we have focused on using logistic regressions to estimate the valuations, we show in Appendix B that symmetry has similar impacts on the valuations estimated through an structural estimation of a cumulative prospect theory model. More generally, constant errors may not be the only errors that arise when subjects make choices in experiments, and other forms of subject response errors may also affect the valuation estimates. Since it is difficult to identify the specific functional forms of mistakes,

¹²Our sample size is a bit smaller than Wu's. After symmetrizing, we again lose half of our data. This turns out to be important for lotteries, where genuine heterogeneity of preferences and additional complexity in valuing them add other layers of uncertainty to the valuation estimates.

this poses a severe challenge for economists to identify people’s preferences from the observed choices. See Zhang (2025) for a thorough discussion using the same data set and a proposed procedure to robustly identify valuations.

Our results also speak to the puzzling finding of Harbaugh, Krause and Vesterlund (2010), who finds the probability weighting patterns to appear only in valuation tasks using the BDM mechanism, but not in binary choice tasks. The BDM mechanism is inherently asymmetric, since it usually restricts the possible valuation to be between the two possible payouts of the lottery.¹³ On the other hand, since the only binary choices they elicit are between the lotteries and their expected values, their binary choices data set is symmetric. As illustrated in this section, this variation in symmetry between their two elicitation methods can explain why they find fourfold pattern only in BDM but not in binary choices.

6 Wu’s Training Teaches More Than Lottery/Mirror Distinctions

So far our results have established that the difference between Wu and Oprea’s result are not driven by a reduction in confusion, but largely differences in the way the choice sets are constructed in the two experiments. In this section we further study the role that training and calculation complexity play in the FFP for mirrors and lotteries.

Wu’s experimental design makes use of \$100 prizes in both training and choice tasks. This may act to reduce FFP in mirrors through two channels, beyond reducing subject confusion. First, it may be that subjects find it easier to calculate the EV of gambles with a prize of \$100. Second, Wu’s training may have taught more than just the distinctions between lottery and mirror: it may also have taught people how to calculate the value of a mirror. Specifically, under Wu’s experimental design, both the lottery $l_{p,100}$ and the mirror $m_{p,100}$ are presented as $100p$ boxes each containing \$100 and $100(1 - p)$ boxes each containing \$0. During the simulation in the training, the subject may simply realize that no calculation

¹³Even when the possible valuation are not restricted, people can still easily realize that the correct valuation has to be between the two possible payouts, and are much less likely to make mistakes by submitting valuations outside of this range.

is needed in order to see how much the mirror pays out, since he can simply look at the number of boxes that contains \$100 and realizes the certain payout is equal to this number. By offering training in this fashion, Wu may be doing more than simply removing confusion about payoff rules.

We have two ways to test this theory. First, we compare the \$25 and \$100 prizes in Experiment 1. If there is something special about the \$100 prize – either due to the fact that it is inherently easier to calculate EVs for these prospects or due to the training – then we should expect to see more FFP in the \$25 prize treatment than the \$100 prize. Figure 9 shows the average relative deviation for both prizes in the main experiment, where relative deviation is defined as deviation as a fraction of the prize. Looking at the mirrors (left panel) as we hypothesized, the fourfold pattern is visibly weaker.¹⁴ On the other hand, for lotteries if anything there is more FFP for the \$100 than for the \$25 prize.

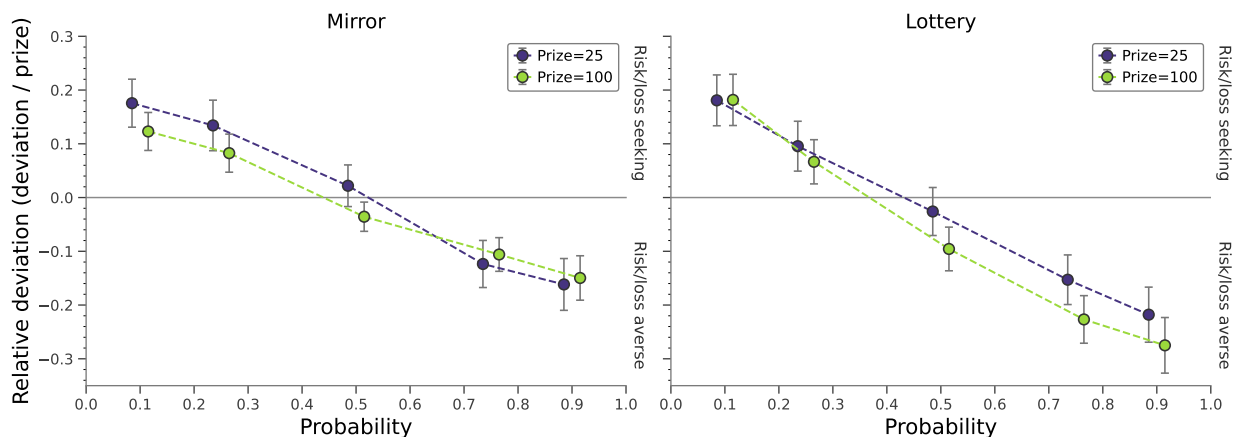


Figure 9: Mean Relative Deviations from Expected Value, under \$100 training and for Both \$25 and \$100 Prizes

These results are supported by regression analysis. As we mention in the previous text, FFP is closely related to the unresponsiveness of valuations with respect to changes in probabilities. To examine the statistical significance of the difference in FFP between prizes \$50 and \$100, we run the following regression separately for mirror and lottery valuations,

¹⁴Since we use the same 26-row MPL to elicit valuations under both prize, many alternative explanations – for example the constant errors we focused on in Section 5 – cannot be responsible for any difference in relative valuations between prizes.

and for each training regime:

$$\text{relative valuation} = \beta_0 + \beta_1 \times \mathbb{1}[\text{prize} = 100] + \beta_2 \times p + \beta_3 \times \mathbb{1}[\text{prize} = 100] \times p, \quad (2)$$

where relative valuation is defined as valuation as a fraction of the prize. This regression measures how responsive valuations are with respect to changes in probabilities, and how this responsiveness changes with different prizes. As shown in table 4, the regression coefficient on the interaction term for mirrors is positive at 0.097 ($p = 0.025$), showing that the responsiveness is significantly higher under prize \$100, and thus FFP is lower under prize \$100.

training	rule	resp 25	resp 100	coef interaction	p interaction
100	mirror	0.552	0.649	0.097	0.025
100	lottery	0.502	0.425	-0.077	0.123
50	mirror	0.778	0.751	-0.026	0.444
50	lottery	0.560	0.618	0.059	0.204

Table 4: Estimated Responsiveness of Valuations and Prizes under Different training Regimes. Table reports the result from regression (2). Columns resp 25 and resp 100 report estimated responsiveness of valuations to prizes under the \$25 and \$100 prizes respectively, for the training regime reported in column training. coef. interaction and p.interaction report the estimated difference in slope between the two training regimes and the associate p values.

Next, we examine how the prize \$100 increases responsiveness and weakens the FFP. We are mainly interested in two types of valuation behaviors. First, the \$100 prize increases the fraction of subjects who value the mirror correctly by 24.8 percentage points ($p < 0.001$). Second, the \$100 prize decreases the fraction of subjects who make constant errors, defined conservatively as those failing to make the payout-maximizing choice even when the payoff difference is greater than 60% of the prize, by 4.1 percentage points ($p = 0.002$). This is consistent with the idea that subjects find it easier to exactly identify the expected value under the \$100 prize and training.

While our results indicate that there may be something special about the \$100 prizes used in Wu’s original design they do not tell us whether the difference is due to the fact that

\$100 prizes are easier to deal with, or because of the match between the training regime and the questions being asked. To differentiate between these explanations, we use the results of the Alternative Training experiment, where we keep the prizes at \$25 and \$100, but change the amount used in training regime to \$50. Under this new training regime, neither of the two prizes, \$25 or \$100, is an exact replication of the training regime. In this environment the difference in strength of the FFP between the two treatments disappears, as can be seen in Figure 10 and Table 4. The interaction terms in Equation (2) is -0.026 ($p = 0.44$). If anything, the responsiveness becomes weaker under prize \$100 than prize \$25. This suggests that the match between the training regime and actual prize does play a role.

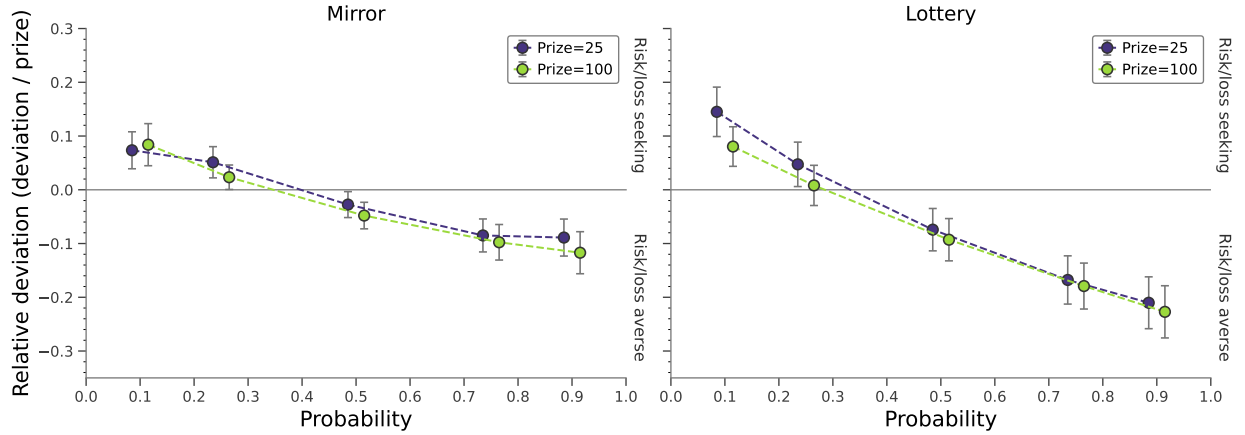


Figure 10: Mean Relative Deviations from Expected Value, under \$50 training and for Both \$25 and \$100 Prizes

It is interesting to note that FFP is less marked for both the \$25 and \$100 prizes in the alternative training experiment relative to the main experiment. It is tempting to conclude that the \$50 training is more effective than the \$100 training in helping subjects to calculate expected values. However, we emphasize that it is dangerous to compare between the two. First, the difficulty of passing the comprehension test varies across the two experiments. When the training uses \$100, it is easy to recognize that, for mirrors, the certain pay is the same as the number of boxes with \$100. In contrast, when the training uses \$50, the same strategy is not possible. Therefore, the subjects who pass the comprehension check and finish the alternative training experiment may be those who pay more attention or possess higher cognitive abilities than those who finish the main experiment. Second, the alternative

training experiment is conducted a few months after the main experiment. Therefore, the subjects recruited in the alternative training experiment may not be fully comparable to the one in the main experiment. As a result, we focus on comparisons between the two prizes within each treatment, but not between the two treatment but fixing a prize.

In summary, by comparing the degree of responsiveness between the two prizes separately for the main treatment and the alternative training experiment, the results suggest that Wu’s training directly teaches people how to calculate the value of a mirror in his original experiment.

7 Discussion and Conclusion

We take our findings as, on the one hand, providing evidence that Oprea’s original results are not driven by confusion, and on the other providing important evidence on the type of behavior that leads to FFP in lotteries and mirrors.

Our first observation is that there we find significant FFP for mirrors and lotteries when we use Wu’s instructions, training regime, comprehension questions, binary choices and a between subject design. Thus the difference between Wu and Oprea’s results cannot be driven by any of these elements reducing confusion - either by reducing ε_{conf}^i or increasing the probability of I^i in our conceptual framework.

Instead we find that if we restrict ourselves to looking at symmetric choices round the EV of a lottery/mirror, then there is a significant reduction in FFP behavior, particularly for mirrors. This is because there is a significant ‘constant’ error component to both lottery and mirror choices, which leads to FFP in the asymmetric, but not symmetric design. Our preferred interpretation is that this represents a component that is common to ε_{calc}^i and μ_{calc}^i . In principle we could assign this behavior to the ‘confusion terms’ ε_{conf}^i and μ_{conf}^i . However, this would imply that such confusion remains an important driver of behavior even under the carefully curated experimental instructions of Wu, which have successfully reduced one obvious measure of confusion (violations of stochastic dominance). As such, we would argue that it is likely an important part of essentially any experimental study into choice under risk, rather than an artifact of Oprea’s design.

Our results do uncover an important difference between mirror and lottery valuations: the latter have additional variation around — and close to — the expected value. This could be due either to subjects’ true utilities $u^i(l_{p,x})$, or lottery-specific errors μ_u^i . This variation leads to some FFP even in the symmetric choice set case.

Finally, we find evidence that the training regime of Wu can act to reduce calculation errors locally, but not generally: the FFP for mirrors is weaker when the training regime and questions use the same prizes than when they are different. If the training was reducing confusion — either by reducing ε_{conf}^i or increasing the probability of I^i — then it should act to reduce FFP even when training and experimental prizes are not matched. We interpret the training regime of Wu as providing very specific information about how to calculate EV for a particular prize, but subjects do not find this information generalizable. This is why it does not have the same effect for mirrors with other prizes, or lotteries with the same prize.

References

- Banki, Daniel, Uri Simonsohn, Robert Walatka, and George Wu.** 2025. “Decisions under Risk Are Decisions under Complexity: Comment.”
- Beauchamp, Jonathan P, Daniel J Benjamin, David I Laibson, and Christopher F Chabris.** 2020. “Measuring and controlling for the compromise effect when estimating risk preference parameters.” *Experimental Economics*, 23(4): 1069–1099.
- Birnbaum, Michael H.** 1992. “Violations of monotonicity and contextual effects in choice-based certainty equivalents.” *Psychological Science*, 3(5): 310–315.
- Blavatskyy, Pavlo R.** 2007. “Stochastic expected utility theory.” *Journal of Risk and uncertainty*, 34(3): 259–286.
- Choi, Syngjoo, Jeongbin Kim, Eungik Lee, and Jungmin Lee.** 2022. “Probability weighting and cognitive ability.” *Management Science*, 68(7): 5201–5215.
- Enke, Benjamin, and Thomas Graeber.** 2023. “Cognitive uncertainty.” *The Quarterly Journal of Economics*, 138(4): 2021–2067.

- Gonzalez, Richard, and George Wu.** 1999. “On the shape of the probability weighting function.” *Cognitive psychology*, 38(1): 129–166.
- Harbaugh, William T, Kate Krause, and Lise Vesterlund.** 2010. “The Fourfold Pattern of Risk Attitudes in Choice and Pricing Tasks.” *The Economic Journal*, 120(545): 595–611.
- Harless, David W, and Colin F Camerer.** 1994. “The predictive utility of generalized expected utility theories.” *Econometrica: Journal of the Econometric Society*, 1251–1289.
- Harrison, Glenn W, Morten Igel Lau, Elisabet E Rutström, and Melonie B Sullivan.** 2005. “Eliciting risk and time preferences using field experiments: Some methodological issues.”
- Loomes, Graham, Peter G Moffatt, and Robert Sugden.** 2002. “A microeconomic test of alternative stochastic theories of risky choice.” *Journal of risk and Uncertainty*, 24(2): 103–130.
- Oprea, Ryan.** 2024. “Decisions under Risk Are Decisions under Complexity.” *American Economic Review*, 114(12): 3789–3811.
- Oprea, Ryan.** 2025. “Initial Reply to Banki, Simonsohn, Walatka and Wu (2025).” , (5182750).
- Wakker, Peter, Ido Erev, and Elke U Weber.** 1994. “Comonotonic independence: The critical test between classical and rank-dependent utility theories.” *Journal of Risk and Uncertainty*, 9(3): 195–230.
- Wu, George.** 2025. “There Is No Smoke with Mirrors When Instructions Are Clear.”
- Zhang, Hanyao.** 2025. “Recovering Preferences from Mistakes: An Auxiliary Task Approach.”

Appendices

A Oprea’s Results Survive Alternative Experimental Designs

Appendix Figure A.1 shows the mean deviation from expected values for lotteries involving \$25 prize, restricting the sample to the first payment rule of each subject. This restricted data set effectively comes from a between-subject treatment design. The graph is virtually the same as Figure 2.

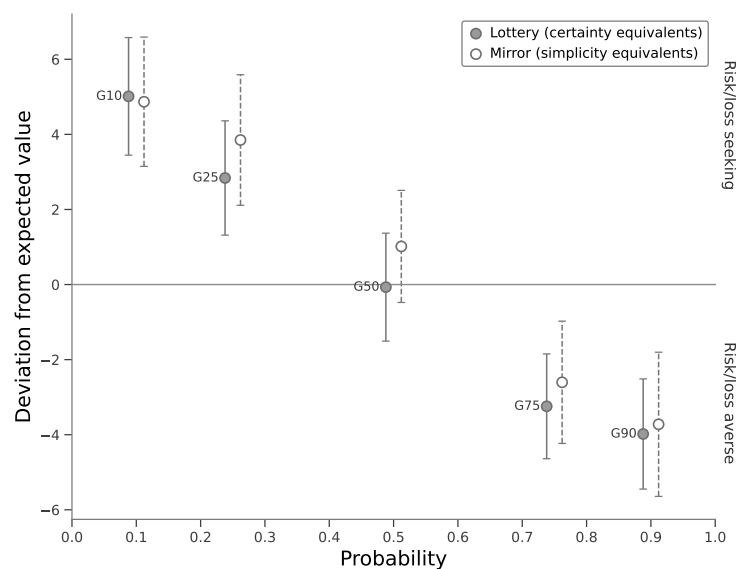


Figure A.1: Mean Deviations from Expected Value in Lotteries (Solid Gray Dots) and Mirrors (Hollow Dots), restricting to the first payment rule of each subject

Appendix Figure A.2 shows the *mean* deviation from expected values for lotteries involving \$25 prize, estimated from the binary choice experiment. We can only estimate an interval of valuation from the binary choice experiment, and the specific procedure through which we generate these estimates are explained at the end of this section. From Appendix Figure A.2, it is clear that Oprea’s headline findings still hold – mirror also exhibits probability weighting patterns despite the fact that it does not entail risk. In the binary choice treatment, we do find that the probability weighting patterns in mirrors become somewhat weaker than in

lotteries.

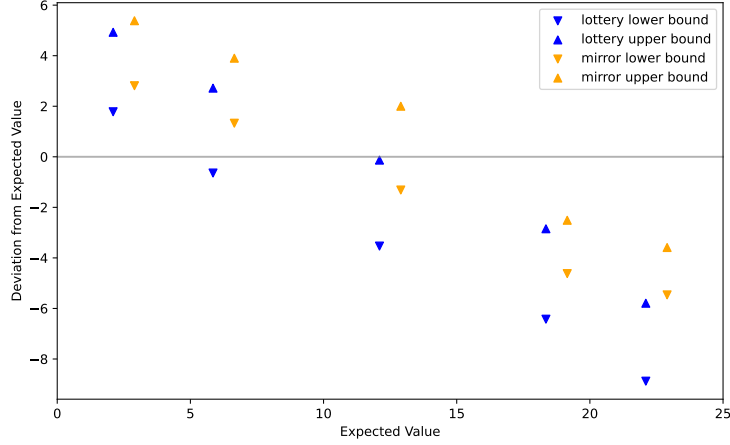


Figure A.2: Upper and Lower Bounds of the Mean Deviations from Expected Value in Lotteries and Mirrors in the Binary Choice Experiment

Below we detail how we construct the estimates for mean valuation from the binary choice data. Readers are free to skip these paragraphs.

For each lottery $l_{p,x}$ or mirror $m_{p,x}$, subjects choose between the lottery/mirror and a certain payment drawn from a fixed grid $\mathcal{M} = \{m_1 < \dots < m_k\} \subset [0, x]$, listed in Table 1. Let $q_p(m_j)$ denote the fraction of subjects who choose the certain payment m_j over the lottery/mirror. If subjects have a well-defined valuation v and choose rationally, and if we have a large enough sample size, then $q_p(m_j)$ estimates the CDF of valuations at m_j : the fraction of subjects with $v \leq m_j$. However, although a CDF must be non-decreasing, in our actual data set q_p may fail to be monotone. Therefore, we cannot simply read off probability masses via first differences of q_p because negative masses would result.

To resolve this, we construct two monotone functions that bracket q_p from above and below, and use them to bound the mean valuation. The key insight is that a CDF that lies *higher* corresponds to a *stochastically smaller* distribution: more probability mass is placed on low values. A higher CDF therefore implies a lower mean. With this in mind, we define:

$$\underline{q}_p(m_j) := \max_{i \leq j} q_p(m_i), \quad \bar{q}_p(m_j) := \min_{i \geq j} q_p(m_i).$$

$\underline{q}_p$ is the smallest non-decreasing function weakly above q_p (a forward running-maximum),

and $\bar{q}_p$ is the largest non-decreasing function weakly below q_p (a backward running-minimum). Because $\underline{q}_p \geq q_p \geq \bar{q}_p$, the distribution induced by $\underline{q}_p$ is *smaller* than the true one in the sense of first order stochastic dominance, and the distribution induced by $\bar{q}_p$ is stochastically *larger*. This is why we call them the *lower* and *upper* bound CDFs, respectively, even though $\underline{q}_p$ is the larger of the two functions: “lower” and “upper” refer to the implied mean valuation, not to the value of the function.

From each of the lower and upper bound CDF, we recover the probability mass on each interval $[m_{j-1}, m_j)$ (with $m_0 := 0$, $m_{k+1} := x$) by taking first differences, appending a tail mass for the interval $[m_k, x]$. We then estimate the mean valuation by assigning every subject in an interval a single representative value:

$$\begin{aligned}\underline{E}[v] &:= \sum_{j=1}^{k+1} m_{j-1} [\underline{q}_p(m_j) - \underline{q}_p(m_{j-1})], \\ \overline{E}[v] &:= \sum_{j=1}^{k+1} m_j [\bar{q}_p(m_j) - \bar{q}_p(m_{j-1})],\end{aligned}$$

where $\underline{q}_p(m_0) = \bar{q}_p(m_0) := 0$ and $\underline{q}_p(m_{k+1}) = \bar{q}_p(m_{k+1}) := 1$. For the lower bound $\underline{E}[v]$, subjects in each interval are assigned the *left* endpoint m_{j-1} ; combined with the fact that $\underline{q}_p$ already places more mass on low-value intervals, this produces a conservative lower estimate of the mean. For the upper bound $\overline{E}[v]$, subjects are assigned the *right* endpoint m_j ; combined with $\bar{q}_p$ placing more mass on high-value intervals, this produces a conservative upper estimate. The interval $[\underline{E}[v], \overline{E}[v]]$ is our identified set for the population mean valuation.

B The Effects of Symmetry on CPT-Estimated Valuations

The analysis in Section 5 uses logistic regression to estimate valuations. Here we show that the same conclusion holds under a structural estimation of a Cumulative Prospect Theory (CPT) model, confirming that the result is not an artifact of the logistic regression approach. We estimate the CPT model for both lottery and mirror. We do not interpret our estimates as a literal combination of a utility function and a probability weighting function, but a unified approach to construct valuations from both the standard and symmetrized MPL data.

CPT model. We model the probability that a subject chooses the lottery $l_{p,x}$ over a certain payment $\$m$ using a CPT utility function with a probit stochastic choice rule:

$$P(\text{choose } l_{p,x} \text{ or } m_{p,x} \text{ over } \$m) = \Phi\left(\frac{U - u(m)}{\sigma}\right), \quad (3)$$

where Φ is the standard normal CDF and $\sigma > 0$ governs choice noise. The utility of the lottery/mirror is

$$U = \pi(p) \cdot u(x), \quad \pi(p) = \frac{p^\gamma}{[p^\gamma + (1-p)^\gamma]^{1/\gamma}}, \quad u(x) = x^\alpha$$

and the utility of the certain payment is $u(m) = m^\alpha$. The parameter α captures the curvature of the value function, and γ captures the curvature of the probability weighting function. When $\gamma < 1$, the weighting function has an inverse-S shape, generating the fourfold pattern. When $\gamma = 1$, probability weighting is linear and the model reduces to expected utility. The valuation implied by the estimated parameters is

$$\widehat{\text{CE}} = [\pi(p) \cdot x^\alpha]^{1/\alpha}.$$

As in Section 5, we first break down an MPL into binary choice, and then estimate the CPT model using the binary choices. We estimate (α, γ, σ) by maximum likelihood.

Results. We apply this model to our MPL data with prize \$25, separately for the standard (asymmetric) MPL and the symmetrized MPL defined in Section 5. Appendix Figure B.1 plots the CPT-estimated deviations of the certainty equivalent from the expected value.

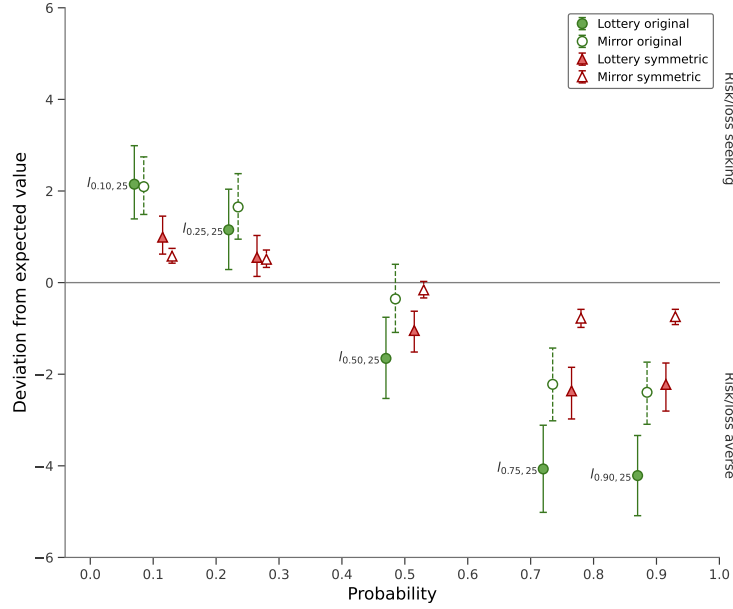


Figure B.1: CPT-estimated valuations, standard MPL versus symmetrized MPL

Note: Valuations are certainty equivalents implied by CPT parameters estimated separately for each condition. The confidence intervals are constructed by estimating the CPT model repeatedly using 1000 resampled subject-level bootstrap data sets.

The qualitative pattern replicates the logistic regression findings. Under the standard MPL, the estimated γ is well below one for both lotteries ($\hat{\gamma} = 0.63$) and mirrors ($\hat{\gamma} = 0.72$), and the average normalized deviations are mostly comparable in magnitude (\$2.65 and \$1.74, respectively). After symmetrization, the mirror γ rises substantially to 0.89, and its average normalized deviation falls to \$0.55, while the lottery γ increases only modestly to 0.76 with average normalized deviation \$1.43. Thus, the CPT estimates confirm that choice set asymmetry can lead to fourfold patterns in mirrors in this setting, and that symmetrization disproportionately attenuates this pattern in mirrors relative to lotteries.

C Experimental Instructions

C.1 Instructions for the Main Experiment and the Alternative Training Experiment

In this study, you will face 10 choice tasks. At the end of the study, if you are randomly selected to receive a bonus payment, then we will randomly select one of the 10 tasks, and we will determine your bonus payment based on your responses in this one randomly selected task.

Note: Each of the decisions you make today may determine your bonus payment. Therefore, it is in your best interest to answer every decision carefully and in a way that reflects what you'd truly prefer.

Next

Boxes paying out money

All choice tasks in this study will involve a set of boxes paying out money according to a payment rule.

Boxes paying out money

You will be facing 100 boxes that pay out money according to a payment rule.

Example:

30 Boxes	70 Boxes
\$100	\$0

Payment rule: You will receive the **average amount of money** in the set of boxes **with certainty**. In this example, you would receive \$30, obtained by $(\$100 \cdot 30 + \$0 \cdot 70) / 100 = \$30$.

Next

3 simulations are provided

Simulated Payment from the Boxes

Example:

75 Boxes	25 Boxes
\$100	\$0

To help you better understand the possible outcomes, we simulate 10 random outcomes from the boxes above.

- \$75
- \$75
- \$75
- \$75
- \$75
- \$75
- \$75
- \$75
- \$75
- \$75

Next

Next Step: Comprehension Check

You'll be asked a few questions to make sure you understand the task. To pass the check, you need to answer 4 out of 5 questions correctly.

If you answer fewer than 4 questions correctly, you will be screened out of the study. You will not be paid for your participation in this case.

If you need to review the instructions, a link will be available for quick access.

Feel free to proceed when you're ready!

Next

5 comprehension questions are asked

Comprehension Question

Imagine you have the following option. If we simulate this option just like what we did, what possible outcomes might you see? Choose all that apply.

[Review Instructions](#)

30 boxes	70 boxes
\$100	\$0

☐ \$0 ☐ \$10 ☐ \$30 ☐ \$50 ☐ \$70 ☐ \$90 ☐ \$100

Choice Tasks

Each choice task will involve one or several decisions between a set of boxes called Option A and a certain payment called Option B. For each decision, all you have to do is decide whether you prefer Option A or Option B.

Your decisions in the choice tasks will be presented in a table in which each row represents a separate decision. In the example that you can find in the bottom of this page, the decision in row 1 involves the choice between Option A (getting paid according to the boxes) and Option B (a certain payment of **\$0**). The decision in row 2 involves a choice between the same Option A, but now Option B gives a 100% chance of receiving **\$1**. In row 3, Option B gives a 100% chance of receiving **\$2**, and so on.

The amount of money in Option B will change in each row. We will put the money amount in **bold** to help you see what changes.

Option A will stay the same in each row of a given choice task.

Example of a choice task

Option A is being paid according to the boxes:

100 Boxes	0 Boxes
\$10.50	\$0

Decision	Option A	Option B
1	Being paid according to the boxes	\$0.00 for sure
2	Being paid according to the boxes	\$1.00 for sure
3	Being paid according to the boxes	\$2.00 for sure
4	Being paid according to the boxes	\$3.00 for sure
5	Being paid according to the boxes	\$4.00 for sure
6	Being paid according to the boxes	\$5.00 for sure
7	Being paid according to the boxes	\$6.00 for sure
8	Being paid according to the boxes	\$7.00 for sure
9	Being paid according to the boxes	\$8.00 for sure
10	Being paid according to the boxes	\$9.00 for sure
11	Being paid according to the boxes	\$10.00 for sure
12	Being paid according to the boxes	\$11.00 for sure
13	Being paid according to the boxes	\$12.00 for sure
14	Being paid according to the boxes	\$13.00 for sure
15	Being paid according to the boxes	\$14.00 for sure
16	Being paid according to the boxes	\$15.00 for sure

Next

Choice Tasks: Example

To give you an example of how these choice tasks work, consider the hypothetical task below. Option A is 100 boxes, each paying out \$10.50. Option B gives a sure payment that varies from \$0 (in the first row) to \$15 (in the last row).

In this hypothetical decision, if you prefer more money to less, you should prefer Option A (paying out \$10.50 for sure) for the first 11 rows. Starting from row 12 (100% chance of \$11), you should prefer Option B.

To avoid having to make a selection in each row, you need only click two times: once for Option A and once for Option B. Once you click on Option A in a particular row, all rows above that row will populate to indicate that you choose Option A. Once you click a row in Option B, all rows below that row will populate to indicate that you choose Option B.

In the example below, in the 11th row, you would click Option A (being paid according to the boxes), instead of Option B ("10 for sure"). In contrast, in the 12th row, you would click Option B ("11 for sure") instead of Option A.

If you do what is described above, your choices will be as in the table below.

IMPORTANT: In order to progress, you have to make a selection for every row.

An hypothetical example of the choices you will be making

Option A is being paid according to the boxes:

100 Boxes	0 Boxes
\$10.50	\$0

Your choice will be the following if you prefer more money to less:

Decision	Option A	Option B
1	Being paid according to the boxes	\$0.00 for sure
2	Being paid according to the boxes	\$1.00 for sure
3	Being paid according to the boxes	\$2.00 for sure
4	Being paid according to the boxes	\$3.00 for sure
5	Being paid according to the boxes	\$4.00 for sure
6	Being paid according to the boxes	\$5.00 for sure
7	Being paid according to the boxes	\$6.00 for sure
8	Being paid according to the boxes	\$7.00 for sure
9	Being paid according to the boxes	\$8.00 for sure
10	Being paid according to the boxes	\$9.00 for sure
11	Being paid according to the boxes	\$10.00 for sure
12	Being paid according to the boxes	\$11.00 for sure
13	Being paid according to the boxes	\$12.00 for sure
14	Being paid according to the boxes	\$13.00 for sure
15	Being paid according to the boxes	\$14.00 for sure
16	Being paid according to the boxes	\$15.00 for sure

Next

Choice Tasks: Practice

To make sure you are comfortable with how the choice tables work, please complete the following practice choice.

1. How to select Option A in all rows
2. How to select Option B in all rows
3. How to change your answer after selecting

As in the main decisions, you will not be able to click to the next page until you have made a valid selection. To make a valid selection, you have to make a selection for every row. That is, your choices must either "switch" from Option A to Option B (clicking once on a row in Option A and once on the subsequent row in Option B to proceed), or you can click Option A in all rows, or Option B in all rows.

If the page is not letting you proceed, you have not made a valid selection. Please make sure you click on Option A in the last row where you would want to select Option A, and then click on Option B in the first row you would want to select Option B (the subsequent row after selecting Option A).

Decision	Option A	Option B
1	Being paid according to the boxes	\$0.00 for sure
2	Being paid according to the boxes	\$1.00 for sure
3	Being paid according to the boxes	\$2.00 for sure
4	Being paid according to the boxes	\$3.00 for sure
5	Being paid according to the boxes	\$4.00 for sure
6	Being paid according to the boxes	\$5.00 for sure
7	Being paid according to the boxes	\$6.00 for sure
8	Being paid according to the boxes	\$7.00 for sure
9	Being paid according to the boxes	\$8.00 for sure
10	Being paid according to the boxes	\$9.00 for sure
11	Being paid according to the boxes	\$10.00 for sure
12	Being paid according to the boxes	\$11.00 for sure
13	Being paid according to the boxes	\$12.00 for sure
14	Being paid according to the boxes	\$13.00 for sure
15	Being paid according to the boxes	\$14.00 for sure
16	Being paid according to the boxes	\$15.00 for sure

Note that you can still change your selection once the table has been auto-populated. To do so, simply click the new row in which you would want to start choosing Option B. Once you have made your final decision, you will be able to proceed to your next choice.

Confirm Choices

Payment

Your bonus payment

After you complete all 10 choice tasks, you will get a participation fee of \$3.00. In addition, you may receive a bonus payment based on your choices. Here is how we will determine your bonus payment.

First, the computer will randomly select a number between 1 and 100, with each number equally likely. If the number drawn is a number between 1 and 4 (inclusive), you have been chosen to receive a possible bonus payment. In other words, you have 4% chance of receiving a bonus payment.

If you are chosen, the computer will randomly select a **choice task** that determines your payment. Then, the computer will randomly select a **decision** from that choice task (one of the rows) by drawing another random number, again with each row equally likely. This will be the **decision-that-counts**. We will determine your payment based on whatever option you chose in the decision-that-counts.

An example

For example, imagine that you had filled out the decision screen below as indicated in the bottom of the screen, and imagine further that the decision-that-counts turned out to be the third row of this decision screen (being paid according to the boxes vs. \$40 for sure).

In this decision, you chose Option A (being paid according to the boxes) over Option B (\$40 for sure). In this case, you would be paid according to the boxes if you are one of the randomly selected participants who will receive a bonus payment. [Review How Boxes Work](#)

If instead you had picked Option B, you will be paid \$40 for sure.

Note, these bonus payments are in addition to your completion payment.

Your Hypothetical Choices

Option A is being paid according to the boxes:

50 Boxes	50 Boxes
\$100	\$0

Decision	Option A	Option B
1	Being paid according to the boxes	\$0.00 for sure
2	Being paid according to the boxes	\$20.00 for sure
3	Being paid according to the boxes	\$40.00 for sure
4	Being paid according to the boxes	\$60.00 for sure
5	Being paid according to the boxes	\$80.00 for sure
6	Being paid according to the boxes	\$100.00 for sure

Next

The main choice task interface

1/10

For each row in the table below, please choose whether you prefer Option A or Option B by clicking on your preferred option.

Option A is being paid according to the boxes:

[Review Instructions](#)

25 Boxes	75 Boxes
\$25	\$0

Option B is a certain amount of money that changes with each decision.

Decision	Option A	Option B
1	Being paid according to the boxes	\$0.00 for sure
2	Being paid according to the boxes	\$1.00 for sure
3	Being paid according to the boxes	\$2.00 for sure
4	Being paid according to the boxes	\$3.00 for sure
5	Being paid according to the boxes	\$4.00 for sure
6	Being paid according to the boxes	\$5.00 for sure
7	Being paid according to the boxes	\$6.00 for sure
8	Being paid according to the boxes	\$7.00 for sure
9	Being paid according to the boxes	\$8.00 for sure
10	Being paid according to the boxes	\$9.00 for sure
11	Being paid according to the boxes	\$10.00 for sure
12	Being paid according to the boxes	\$11.00 for sure
13	Being paid according to the boxes	\$12.00 for sure
14	Being paid according to the boxes	\$13.00 for sure
15	Being paid according to the boxes	\$14.00 for sure
16	Being paid according to the boxes	\$15.00 for sure
17	Being paid according to the boxes	\$16.00 for sure
18	Being paid according to the boxes	\$17.00 for sure
19	Being paid according to the boxes	\$18.00 for sure
20	Being paid according to the boxes	\$19.00 for sure
21	Being paid according to the boxes	\$20.00 for sure
22	Being paid according to the boxes	\$21.00 for sure
23	Being paid according to the boxes	\$22.00 for sure
24	Being paid according to the boxes	\$23.00 for sure
25	Being paid according to the boxes	\$24.00 for sure
26	Being paid according to the boxes	\$25.00 for sure

Confirm Choices

After 5 tasks, the subjects enter the second payment rule and are shown information about the second payment rule like below

Boxes paying out money

In the upcoming choice tasks, the **payment rule has changed**. Please review the new payment rule below.

Boxes paying out money

You will be facing 100 boxes that pay out money according to a payment rule.

Example:

30 Boxes	70 Boxes
\$100	\$0

Payment rule: The computer will calculate the outcome by **randomly picking** one of the boxes. The amount of money in that box (in this example, could be \$0 or \$100) will be your final result.

Next

The simulation and comprehension checks will appear again, but this time for the new payment rule.

C.2 Instructions for the Binary Choice Experiment

The instructions are largely the same as those in Section C.1, and we only show the parts where they differ.

The first two pages of the binary choice experiment differ from the main experiment by tailoring to binary choices.

In this study, you will face 20 choice tasks.

After you complete all 20 choice tasks, you will get a participation fee of \$3.00. In addition, 4 out of 100 participants will be randomly selected to receive a bonus payment.

If you are randomly selected to receive a bonus payment, then we will randomly select one of the 20 choice tasks, and we will determine your bonus payment based on your responses in this one randomly selected task.

Note: Each of the decisions you make today may determine your bonus payment. Therefore, it is in your best interest to answer every decision carefully and in a way that reflects what you'd truly prefer.

Next

Boxes paying out money

The next set of choice tasks in this study will involve two sets of boxes paying out money according to a payment rule.

Boxes paying out money

You will be making choices between two options: A and B. Each option contains 100 boxes.

You may choose the option that leads to the outcome that you prefer.

Example:

Option A

30 Boxes	70 Boxes
\$100	\$0

Option B

100 Boxes
\$60

How is your outcome determined by the option you choose?

Payment rule: You will receive the **average amount of money** in the boxes of the option you choose, **with certainty**. For example,

- If you choose Option A, you would receive \$30 for sure, obtained by $(\$100 \cdot 30 + \$0 \cdot 70) / 100 = \$30$.
- If you choose Option B, you would receive \$60 for sure, obtained by $\$60 \cdot 100 / 100 = \60 .

Next

Then, the subject works through the simulation and comprehensions exactly as in Sec-

tion C.1 (the pages up to the one titled *Choice Tasks* in Section C.1).

After the instructions the subject begins the main task, with the interface shown below. The left-right placing of the mirror and the sure payment is randomized across trials.

1/20

Please select which Option (A or B) you'd prefer.

Review Instructions

☐ Option A

☐ Option B

50 Boxes	50 Boxes
\$25	\$0

100 Boxes
\$15

Next

After 10 tasks, the subjects enter the second payment rule and are shown information about the second payment rule like below.

Boxes paying out money

In the upcoming choice tasks, the **payment rule has changed**. Please review the new payment rule below.

Boxes paying out money

You will be making choices between two options: A and B. Each option contains 100 boxes.

You may choose the option that leads to the outcome that you prefer.

Example:

Option A

30 Boxes	70 Boxes
\$100	\$0

Option B

100 Boxes
\$60

How is your outcome determined by the option you choose?

Payment rule: The computer will calculate the outcome by **randomly picking** one of the boxes from the option you choose. The amount of money in that box will be your final result.

Next

The simulation and comprehension checks will appear again, but this time for the new payment rule.

44